

# Dr. Kathryn I. Wheeler

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## EDUCATION

### Boston University

Boston, MA

*Doctor of Philosophy in Earth and Environment* (GPA: 3.95)

Jan. 2023

*Certificate in Biogeosciences*

Dissertation: "Cold-deciduous broadleaf phenology: monitoring using a geostationary satellite and predicting using trigger-less dynamic models"

Advised By: Dr. Michael Dietze

Academic Awards: National Science Foundation Graduate Research Fellowship and Dean's Fellowship

Relevant Courses: Remote sensing, multivariate statistics, Bayesian statistics, ecological forecasting, terrestrial carbon

### University of Delaware

Newark, DE

*Bachelor of Science in Environmental Science* (GPA: 3.95)

May 2017

*Honors with Distinction, summa cum laude*

Academic Awards: Full four-year merit scholarship, NOAA Hollings Scholarship, UD American Association of University

Professors Outstanding Senior, and Outstanding Senior in Environmental Science

Relevant Courses: GIS, soil science, plant physiology, ecosystem ecology, computer science, data structures

## RELEVANT EXPERIENCE

### Industrial Economics, Incorporated

Remote and Cambridge, MA

*Environmental Consulting Associate*

June 2023 – Present

- Used R, Python, and ArcGIS Pro to process and analyze spatial-temporal environmental data to assist with Natural Resource Damage Assessments at several Superfund sites
- Improved efficiency of a Rshiny app to display spatial data to clients by 5x
- Performed advanced statistical models of spatial and spatial-temporal Inverse Distance Weighting, distributed lag for time series, PCA, mixed effects, generalized estimating equations, and non-negative matrix factorization to estimate contaminant impacts on fish populations
- Translated scientific findings to clients with limited technical knowledge through reports and presentations
- Identified patterns in complex datasets using machine learning (decision trees and k-means clustering)
- Saved 120+ hours by using Python to automate running a client-preferred point-and-click application

### Massachusetts Institute of Technology

Cambridge, MA

*NOAA Climate and Global Change Postdoctoral Fellow*

Sep. 2022 – June 2023

- Funded through a highly competitive 2-year fellowship (\$200k) selected through a written research proposal
- Utilized Google Earth Engine and MODIS data to assess drought impacts on tropical forest biomass
- Used LiDAR and eddy covariance flux data to investigate relationship between forest structure and productivity
- Led a multidisciplinary team of 43 to conduct ecological forecasts of tree leaf growth timing
- Coded Bayesian and random forest machine learning models to analyze impacts of soil and biota on leaf seasonality

### Boston University

Boston, MA

*Research Assistant*

Sep. 2017 – Aug. 2022

- Created novel Bayesian statistical and mechanistic models in R to predict tree leaf seasonality
- Conducted field work at Harvard Forest to investigate the link between senescence and photosynthesis
- Developed automation pipeline to download, clean, organize, and condense 30 TB of satellite data from MODIS and GOES using a high-performance distributed system

*Teaching Fellow and Lab Instructor*

Jan. 2022 – May 2022

- Led weekly lab based in R for 30 students in an environmental modeling course

### University of Delaware

Newark, DE

- Analyzed water quality data using MATLAB to fit advanced statistical and neural network models to available data
- Collected and analyzed 1,700 hyperspectral leaf measurements using a spectrophotometer and Python
- Presented research at AGU and won two student presentation awards in hydrology and biogeoscience (top 5%)

- Provided teaching assistance for a weekly lab based in Python and MATLAB

## PUBLICATIONS

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1. **Wheeler** et al. (2024). "Predicting spring phenology in deciduous broadleaf forests: NEON Phenology Forecasting Community Challenge." *Agr Forest Meteorol* 345: 109810.
  2. Thomas et al.\* (2023). "The NEON Ecological Forecasting Challenge." *Front Ecol Evol* 21(3): 112–113.
  3. Halpern et al.\* (2023) "Priorities for synthesis in ecology and environmental science." *Ecosphere* 14(1):e4342.
  4. **Wheeler** & Dietze (2021). "Improving the monitoring of deciduous broadleaf phenology using the Geostationary Operational Environmental Satellite (GOES) 16 and 17." *Biogeosciences* 18: 1971-1985.
  5. Iida\* & **Wheeler**\* et al. (2021). "Canopy structure metrics governing stemflow funneling differs between leafed and leafless states: Insights from a large-scale rainfall simulator". *Hydrol Processes* 35:e14294.
- \*These authors contributed equally to this paper**
6. Woelmer et al.\* (2021). "10 Simple Rules for training yourself in an emerging field" *PLoS Comput Biol* 17(10): e1009440. <https://doi.org/10.1371/journal.pcbi.1009440>
  7. Rollinson et al.\* (2021) "Working across space and time: nonstationarity in ecological research and application." *Front Ecol Evol* 19(1): 66-72.
  8. **Wheeler** et al. (2020). "Visible and near-infrared hyperspectral indices explain more variation in lower-crown leaf nitrogen concentrations in autumn than in summer." *Oecologia*. 192:13-27.
- Award: Highlighted Student Paper**
9. **Wheeler** & Dietze (2019). "A statistical model for estimating midday NDVI from the Geostationary Operational Environmental Satellite (GOES) 16 and 17." *Remote Sensing* 11(21):2507.
  10. Dietze et al.\* (2018). "Ecological Forecasting." *Oxford Bibliographies*.
  11. Hudson et al.\* (2018). "American beech leaf-litter leachate chemistry: effects of geography and phenophase." *J Plant Nutr Soil Sci* 181(2):287-295.
  12. **Wheeler** et al. (2017). "Tracking senescence-induced patterns in leaf litter leachate using parallel factor analysis modeling (PARAFAC) and self-organizing maps." *J. Geophys Res Biogeosci* 122(9):2233-2250.
  13. Preprint: **Wheeler** & Dietze (2023). "A trigger may not be necessary to cause senescence in deciduous broadleaf forests." *bioRxiv* 10.1101/2023.06.07.544057.

\*Includes K.I. Wheeler

## SKILLS AND INTERESTS

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**Programming Languages:** R (Expert), Python (Intermediate), MATLAB (Intermediate), Java (Familiar), C++ (Familiar)

**Geospatial Tools:** spatial R packages (Expert: sf, terra, raster), Google Earth Engine (Intermediate), ArcGIS (Intermediate), spatial and spatial-temporal Inverse Distance Weighting

**High-Performance Computing Tools:** Git (expert), shared clusters, parallelization, shell scripting (Unix, Bash), Slurm and Open Grid schedulers

**Data Analysis Knowledge:** Bayesian statistics (MCMC, state space, hierarchical models), frequentist statistics (distributed lag time series, Principal Component Analysis, mixed effects models, generalized estimating equations, non-negative matrix factorization, generalized additive models), machine learning (random forest, self-organizing maps, clustering), power analysis, sensitivity analysis, propagating and analyzing uncertainty, model assessment, data fusion and assimilation (Kalman Filter, Extended Kalman Filter, Ensemble Filters, Particle Filters)

**Domain Knowledge:** remote sensing, ecological forecasting, geospatial analysis, environmental modeling, forest ecology, carbon cycle, data science, AI capabilities (Claude)

**Communication Skills:** proposal writing, technical presentations, teaching, project management, leadership, mentorship, Markdown, Jupyter

**Interests:** running, hiking, cycling, backpacking, and reading